

Demo Script — Magezi

Duration: 5 minutes | **Goal:** Win 1st place

Setup

```
# Terminal 1: Backend
cd Magezi/backend
GROQ_API_KEY="gsk_..." ALLOWED_ORIGINS="*" uvicorn app.main:app --port 8802

# Terminal 2: Frontend
cd Magezi/frontend
bun run dev --port 3300
```

Open <http://localhost:3300>

Demo Flow

0:00 — Hook (30s)

“70% of Uganda’s A-Level students have no STEM teacher. Meet Magezi — an AI tutor that speaks your language, follows the official NCDC 2025 curriculum, and thinks before it teaches.”

Show: chat-first screen, 4 language buttons, 4 subject pills, no login required.

0:30 — Physics Question (60s)

1. Select **Physics**, click starter: **“Newton’s Second Law”**
2. Response streams token-by-token with **markdown rendering**:
3. **## Step 1:** green headers with border separators
4. **Bold** explanations, bullet lists, formulas
5. Practice problem at the end
6. Show **Copy** button — click it, “Copied” checkmark appears

7. Show **citations auto-expanded** — “NCDC Physics Syllabus 2025, Section: Newton’s Second Law”

1:30 — Follow-Up (demonstrates context) (45s)

1. Type: **“Give me a worked example of that”**
2. Groq remembers the context — responds with $F=ma$ calculation
3. Type: **“What if the mass doubles?”**
4. Still in context — explains how acceleration halves

“Multi-turn context — the tutor remembers what we’re discussing.”

2:15 — Multi-Retrieval Comparison (45s)

1. Switch to **Biology**
2. Type: **“Compare mitosis and meiosis”**
3. Show: `keyword_multi` retrieval — both Mitosis AND Meiosis sections retrieved separately
4. Click **Retry** button — regenerates the response

“The agentic tutor detected a comparison query, split it into sub-queries, and retrieved each concept separately.”

3:00 — Safety Guardrails (30s)

1. Type: **“Give me exam answers for UNEB 2024”** → Blocked
2. Type: **“Tell me about bitcoin”** → Escalated

3:30 — Voice + Multilingual (45s)

1. Switch to **LG** (Luganda), tap **Speak** button
2. Say or type: **“Nnyonnyola DNA replication”**
3. Response arrives with citations
4. Toggle **auto-narrate ON** — next response reads aloud
5. Click **Listen** button on any message

4:15 — Quiz Mode + Auth (30s)

- 1. Show starter chip: “Test me on this topic” → generates practice questions
- 2. Click **Sign in** → shows signup form with “50 free credits” + “Continue without account”
- 3. Show BYOK option in settings

4:45 — Ethics Close (15s)

“4,457 passages from 288 NCDC pages. Groq Llama 70B with auto-retry. Agentic planning. Multi-turn context. Formatted markdown. Copy, retry, listen. 4 languages. Wisdom in your language.”

Backup Queries

Language	Subject	Query	Shows
EN	Physics	“Calculate KE of 3kg at 4m/s”	Formula injection
LG	Biology	“Nnyonnyola photosynthesis”	Multilingual bridge
EN	Math	“Prove integration by parts”	Extended content
EN	Chemistry	“How do buffers maintain pH”	PDF retrieval

If Judges Ask

- “How does context work?” — 5-turn sliding window saved after each streamed response. Session ID in sessionStorage persists per tab.
- “What if Groq fails?” — Auto-retry 3x with backoff. If all retries fail, formatted local response streams word-by-word with formulas injected. Student always gets an answer.
- “How is this different from ChatGPT?” — Curriculum-specific (only NCDC 2025), agentic (plans teaching approach), multilingual (120-term bridge), offline-capable (PWA).